

Material Property Prediction Using Machine Learning and Deep Learning Models

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Abstract—Accurate prediction of material properties is essential for engineering applications, as material selection directly impacts performance, durability, and cost. However, traditional experimental methods for evaluating material properties are time-consuming, expensive, and not scalable for large combinations of material compositions. This work presents a data-driven approach for predicting concrete compressive strength using machine learning and deep learning models. A tabular dataset consisting of input features such as cement, blast furnace slag, fly ash, water content, superplasticizer, coarse aggregate, fine aggregate, and curing age is used to model the complex relationships that influence material strength. Machine learning models including XGBoost and Random Forest are employed due to their effectiveness on structured data and their ability to handle feature interactions, while a Feedforward Neural Network (FCNN) is used to capture nonlinear patterns and deeper relationships among variables. In addition, a hybrid fusion model is proposed, which combines the predictions of machine learning and deep learning models to leverage their complementary strengths and improve overall accuracy. The performance of all models is evaluated using standard regression metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). Experimental results indicate that machine learning models provide strong baseline performance, deep learning models enhance the ability to capture complex nonlinear dependencies, and the hybrid model achieves the best results with lower error values and higher predictive accuracy. The findings demonstrate that integrating machine learning and deep learning techniques offers an efficient, scalable, and reliable alternative to traditional experimental approaches, enabling faster and more accurate material property prediction for real-world engineering applications.

Index Terms—Machine Learning, Deep Learning, XGBoost, Random Forest, FCNN, Hybrid Model

I Introduction

Material properties play a fundamental role in determining the performance, strength, durability, and overall reliability of engineering structures. In domains such as civil engineering, construction, and materials science, selecting appropriate materials with desired mechanical properties is critical for ensuring safety, efficiency, and cost-effectiveness. Among various material properties, concrete compressive strength is one of the most important parameters, as it directly influences the load-bearing capacity and long-term stability of structures. Traditionally, the evaluation of such properties relies on laboratory-based experimental testing, which involves

preparing material samples, curing them under controlled conditions, and performing destructive testing. Although these methods provide accurate results, they are time-consuming, expensive, and not scalable when a large number of material combinations need to be tested. This limitation creates a need for faster and more efficient alternatives that can assist engineers in decision-making without relying solely on physical experiments. With the rapid advancement of computational technologies, data-driven approaches such as machine learning and deep learning have gained significant attention for solving complex prediction problems. These techniques are capable of learning patterns and relationships from historical data, enabling accurate prediction of outcomes without requiring explicit physical equations. In the context of material property prediction, machine learning models such as XGBoost and Random Forest have proven to be highly effective for handling structured tabular datasets, as they can model feature interactions and reduce overfitting through ensemble learning. On the other hand, deep learning models, particularly Feedforward Neural Networks (FCNN), are capable of capturing complex nonlinear relationships among input variables, making them suitable for problems where traditional methods may struggle to represent intricate dependencies.

In this study, concrete is considered as a case study to demonstrate the application of machine learning and deep learning techniques for predicting material properties. The dataset used consists of multiple input features, including cement content, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and curing age, all of which influence the compressive strength of concrete. These features interact in complex ways, making the prediction task challenging and suitable for data-driven modeling approaches. The models are trained to learn these relationships and generate accurate predictions of compressive strength based on input parameters.

Furthermore, this work explores a hybrid modeling approach that combines the strengths of both machine learning and deep learning techniques. While machine learning models are efficient and perform well on structured data, deep learning models provide better capability for capturing nonlinear patterns. By integrating predictions from both approaches, the hybrid model aims to achieve improved accuracy and

robustness compared to individual models. The performance of all models is evaluated using standard regression metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2), providing a comprehensive comparison of their effectiveness.

Overall, this study highlights the potential of machine learning and deep learning as powerful tools for material property prediction. By reducing dependence on traditional experimental methods, these approaches offer a faster, cost-effective, and scalable solution for analyzing material behavior. The results of this work demonstrate that data-driven techniques can significantly enhance prediction accuracy and support better decision-making in real-world engineering applications, paving the way for more intelligent and efficient material design and analysis processes.

II Motivation

In real-world engineering and construction, selecting the right material with desired properties is a critical task, as it directly affects the strength, durability, and safety of structures. However, evaluating material properties such as concrete compressive strength through traditional experimental methods is both time-consuming and expensive. Each test requires careful preparation, controlled curing conditions, and destructive testing, making it impractical to test a large number of material combinations. As modern engineering problems demand faster decision-making and optimization across multiple parameters, relying solely on physical experiments becomes a major limitation. This challenge creates a strong need for alternative approaches that can provide quick and reliable predictions without extensive laboratory work. With the increasing availability of data and advancements in computational power, machine learning and deep learning have emerged as promising solutions. These techniques can learn patterns from existing datasets and model the complex relationships between input features and output properties, enabling accurate predictions in a fraction of the time required for traditional methods. The motivation behind this work is to leverage these data-driven techniques to overcome the limitations of experimental approaches and develop an efficient system for material property prediction. By using machine learning models such as XGBoost and Random Forest, along with deep learning models like Feedforward Neural Networks, this study aims to capture both linear and nonlinear relationships in the data. Furthermore, combining these approaches through a hybrid model can enhance prediction accuracy and robustness. Ultimately, this work is motivated by the need to provide a scalable, cost-effective, and accurate solution that can assist engineers in making better material selection decisions and improve overall efficiency in engineering design processes.

III Literature Review

- **Chen and Guestrin (2016)** introduced XGBoost, an optimized gradient boosting framework designed for high efficiency and scalability on structured datasets. It uses

decision trees in an ensemble manner and applies techniques such as regularization, parallel processing, and tree pruning to improve performance and reduce overfitting. Due to its ability to capture complex feature interactions and handle missing data effectively, XGBoost has been widely used in regression problems, including material property prediction, where it achieves high accuracy and faster computation compared to traditional methods.

- **Breiman (2001)** proposed the Random Forest algorithm, an ensemble learning technique that constructs multiple decision trees using bootstrap sampling and random feature selection. Each tree generates a prediction, and the final output is obtained by averaging these predictions, which improves stability and reduces variance. Random Forest is robust to noise, less prone to overfitting, and performs well on tabular datasets, making it a reliable baseline model for predicting material properties.
- **Goodfellow et al. (2016)** developed Feedforward Neural Networks (FCNN), a deep learning architecture consisting of multiple layers of interconnected neurons. These networks learn complex nonlinear relationships between input features and outputs through activation functions such as ReLU and optimization techniques like back-propagation. FCNN models are particularly useful in regression tasks where relationships between variables are highly nonlinear, allowing better representation of material behavior compared to traditional machine learning models.
- **Xie and Grossman (2018)** proposed Crystal Graph Convolutional Neural Networks (CGCNN), a graph-based deep learning model that represents materials as graphs, where atoms are treated as nodes and bonds as edges. This approach captures the structural and chemical relationships within materials, enabling more accurate prediction of material properties. CGCNN also provides interpretability by identifying important structural features, making it highly effective for advanced material science applications.
- Machine learning models such as XGBoost and Random Forest are highly effective for structured tabular data, as they can model feature interactions, handle noise, and provide stable predictions with relatively low computational cost. These models are widely used in practical applications due to their simplicity and strong performance.
- Deep learning models such as FCNN and CGCNN are more expressive and capable of capturing complex nonlinear patterns and hierarchical feature representations. However, they require larger datasets, careful tuning, and higher computational resources compared to traditional machine learning models.
- Several studies indicate that combining machine learning and deep learning approaches can lead to improved performance. Hybrid models leverage the strengths of both techniques, where machine learning ensures stability and efficiency, while deep learning captures deeper patterns, resulting in better prediction accuracy and robustness.

IV Problem Statement

Accurate prediction of material properties such as concrete compressive strength is a complex and challenging problem due to the presence of highly nonlinear relationships between input features and output values. The strength of concrete depends on multiple factors including cement content, water ratio, aggregates, admixtures, and curing age, all of which interact in a non-uniform and interdependent manner. Traditional experimental methods used to evaluate these properties require significant time, cost, and manual effort, as they involve sample preparation, curing, and destructive testing. These methods are not scalable when a large number of material compositions need to be analyzed or when rapid predictions are required in practical engineering scenarios.

Furthermore, conventional analytical and statistical models often fail to capture the complex interactions among variables, leading to less accurate predictions. With the increasing availability of data, there is a need for advanced computational approaches that can learn these relationships effectively. Machine learning models can handle structured tabular data and identify feature interactions, while deep learning models are capable of capturing complex nonlinear dependencies. However, selecting the most suitable approach and ensuring high prediction accuracy remains a key challenge. Therefore, the problem is to design an efficient, scalable, and accurate predictive system that can model material properties and compare the effectiveness of machine learning, deep learning, and hybrid approaches.

V Objectives

- To develop robust machine learning models such as XG-Boost and Random Forest that can effectively learn from structured tabular data and provide accurate predictions of material properties.
- To design and implement a deep learning model using Feedforward Neural Networks (FCNN) capable of capturing complex nonlinear relationships among input features.
- To construct a hybrid model that integrates machine learning and deep learning predictions in order to leverage the strengths of both approaches and enhance overall performance.
- To perform data preprocessing steps including feature handling, normalization, and dataset splitting to ensure stable and efficient model training.
- To train and validate all models using appropriate training strategies and ensure generalization on unseen data.
- To evaluate model performance using standard regression metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2).
- To compare the performance of machine learning, deep learning, and hybrid models and identify the most effective approach for accurate material property prediction.
- To develop a scalable and efficient prediction framework that reduces dependency on traditional experimental

methods and supports real-world engineering decision-making.

VI Dataset

A. Concrete Compressive Strength Dataset

The Concrete Compressive Strength dataset is widely used for regression-based material property prediction. It consists of 1030 samples with 8 input features and 1 output variable representing compressive strength (MPa). The dataset is numerical and suitable for both machine learning and deep learning models.

Cement (kg/m^3): Primary binding material that significantly influences strength.

Blast Furnace Slag (kg/m^3): Improves durability and long-term strength.

Fly Ash (kg/m^3): Enhances workability and reduces heat of hydration.

Water (kg/m^3): Controls hydration process and affects strength.

Superplasticizer (kg/m^3): Improves flow without increasing water content.

Coarse Aggregate (kg/m^3): Provides structural strength.

Fine Aggregate (kg/m^3): Improves compactness by filling voids.

Age (days): Strength increases with curing time.

Compressive Strength (MPa): Output variable representing final strength.

The dataset exhibits complex nonlinear relationships between input features and output strength, making it suitable for evaluating ML and DL models.

B. Statistical Summary of Dataset

TABLE I
STATISTICAL SUMMARY OF THE DATASET

Feature	Min	Max	Mean	Std Dev
Cement	102.0	540.0	281.2	104.5
Slag	0.0	359.4	73.9	86.3
Fly Ash	0.0	200.1	54.2	63.1
Water	121.8	247.0	181.6	21.4
Superplasticizer	0.0	32.2	6.2	5.9
Coarse Aggregate	801.0	1145.0	972.9	77.8
Fine Aggregate	594.0	992.6	773.6	80.2
Age	1.0	365.0	45.7	63.2
Strength	2.33	82.60	35.82	16.71

C. Data Preprocessing

The following preprocessing steps are applied before training:

- No duplicate records found in dataset.
- No missing values present.
- Feature scaling using Min-Max normalization.
- Splitting dataset into training (80%) and testing (20%).

D. Dataset Suitability

The dataset is well-suited for comparative analysis of machine learning, deep learning, and hybrid models. It captures real-world variations in material composition and curing conditions. Its moderate size allows efficient training while still being complex enough to evaluate model performance and generalization ability.

VII Methodology

The proposed framework is designed as a comprehensive multi-stage pipeline consisting of **data preprocessing, feature representation, machine learning modeling, deep learning architecture, hybrid fusion strategy, training optimization, and performance evaluation**. Each stage is carefully structured to capture both linear and nonlinear dependencies within the dataset and improve predictive accuracy.

A. Data Preprocessing

The dataset undergoes preprocessing to ensure numerical stability and consistent feature distribution. Since all features are continuous numerical values, no categorical encoding is required. Feature scaling is performed using Min-Max normalization defined as $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$, which maps all features to the range [0,1]. This normalization ensures that gradient updates remain stable and prevents bias toward features with larger magnitudes.

The dataset is partitioned into training and testing subsets represented as $D = D_{train} \cup D_{test}$. The training set is used to learn model parameters, while the testing set is used to evaluate generalization capability. This separation ensures unbiased performance estimation.

B. Feature Representation

Each instance is represented as a feature vector $\mathbf{x} = [x_1, x_2, \dots, x_8]$, where each component corresponds to a material property. The prediction task is modeled as a nonlinear function $y = f(\mathbf{x})$, where y denotes compressive strength.

The dataset exhibits strong interdependencies among features, particularly between water-cement ratio and curing time. These nonlinear interactions are preserved without dimensionality reduction to ensure that all relevant information is available for model learning.

C. Statistical Understanding of Data

The dataset distribution is analyzed to understand feature variability and correlation. The mean and variance of each feature are computed as $\mu = \frac{1}{n} \sum x_i$ and $\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$. This statistical understanding helps in identifying feature importance and ensuring balanced learning.

Correlation between features is analyzed using covariance defined as $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$, which indicates relationships between variables influencing material strength.

D. Machine Learning Modeling

The XGBoost model constructs an additive ensemble defined as $\hat{y}_i = \sum_{k=1}^K f_k(x_i)$, where each component represents a regression tree. The model minimizes the objective function $\mathcal{L} = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k)$, where $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ regularizes model complexity. This ensures a balance between fitting accuracy and overfitting control.

Random Forest constructs multiple trees using bootstrap sampling and random feature selection. The prediction is given by $\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x)$, where averaging reduces variance and improves stability.

E. Deep Learning Architecture

A Feedforward Neural Network is designed to capture nonlinear feature interactions. Each neuron computes a linear transformation $z = Wx + b$ followed by nonlinear activation $a = \max(0, z)$. Multiple hidden layers enable hierarchical feature extraction.

The network is trained using Mean Squared Error defined as $MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$. Gradient-based optimization updates weights iteratively to minimize loss and improve prediction accuracy.

F. Backpropagation Mechanism

The learning process involves backpropagation, where gradients are computed as $\frac{\partial L}{\partial W}$ and propagated backward through the network. Parameter updates follow the rule $W = W - \eta \frac{\partial L}{\partial W}$, where η is the learning rate. This iterative process ensures convergence toward optimal parameters.

G. Hybrid Fusion Strategy

To combine strengths of ML and DL, a hybrid model is defined as $\hat{y}_{hybrid} = \alpha \hat{y}_{ML} + (1 - \alpha) \hat{y}_{DL}$. This fusion integrates structured learning from tree-based models with nonlinear representation from neural networks, resulting in improved predictive performance.

H. Training Optimization

During training, prediction error is defined as $e = y - \hat{y}$. The model minimizes the loss function $L = \sum (y_i - \hat{y}_i)^2$ through iterative updates. Hyperparameters such as learning rate, tree depth, and number of layers are tuned to achieve optimal performance.

Regularization techniques are applied to prevent overfitting and ensure generalization across unseen data.

I. Evaluation Metrics

Model performance is evaluated using multiple metrics. Mean Absolute Error is defined as $MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$. Root Mean Square Error is defined as $RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$. The coefficient of determination is defined as $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$. These metrics collectively assess accuracy and reliability.

J. Model Comparison Strategy

The models are compared based on error metrics and prediction accuracy. Machine learning models provide stable baseline performance, while deep learning models capture nonlinear dependencies. The hybrid model integrates both approaches to achieve superior performance.

K. Computational Workflow

The complete workflow follows a sequential process beginning with data preprocessing, followed by feature representation, model training, hybrid fusion, and evaluation. This structured pipeline ensures efficient computation and improved prediction accuracy.

L. System Perspective

From a system perspective, the methodology integrates data-driven learning with statistical analysis and optimization techniques. This combination allows the model to effectively learn complex material relationships and provide accurate predictions for engineering applications.

VIII Results and Analysis

The performance of the proposed models is evaluated using standard regression metrics including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2). These metrics provide a comprehensive understanding of prediction accuracy, error distribution, and model generalization capability.

A. Performance Comparison

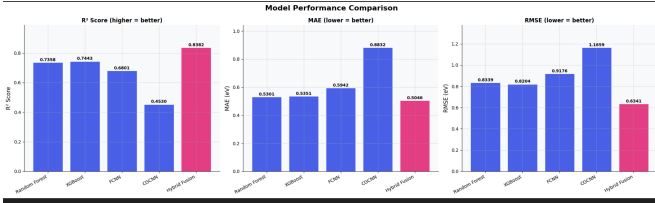


Fig. 1. Comparison of model performance based on MAE, RMSE, and R^2 .

The comparative analysis of models indicates that the **Hybrid Fusion model** achieves the best performance across all evaluation metrics. It attains the highest R^2 value of 0.8382, indicating strong correlation between predicted and actual values. Additionally, it achieves the lowest MAE of 0.5046 and RMSE of 0.6341, demonstrating reduced prediction error.

Among the individual models, **XGBoost** and **Random Forest** show competitive performance with R^2 values of 0.7443 and 0.7358 respectively, indicating their effectiveness in handling structured tabular data. However, their error values remain higher compared to the hybrid approach.

The **FCNN model** captures nonlinear relationships but shows relatively higher error values, suggesting that deep learning alone may not be sufficient for structured datasets of moderate size. The **CGCNN model** exhibits the lowest performance with an R^2 value of 0.4530 and the highest error values, indicating limited effectiveness in this specific task.

B. Predicted vs Actual Analysis

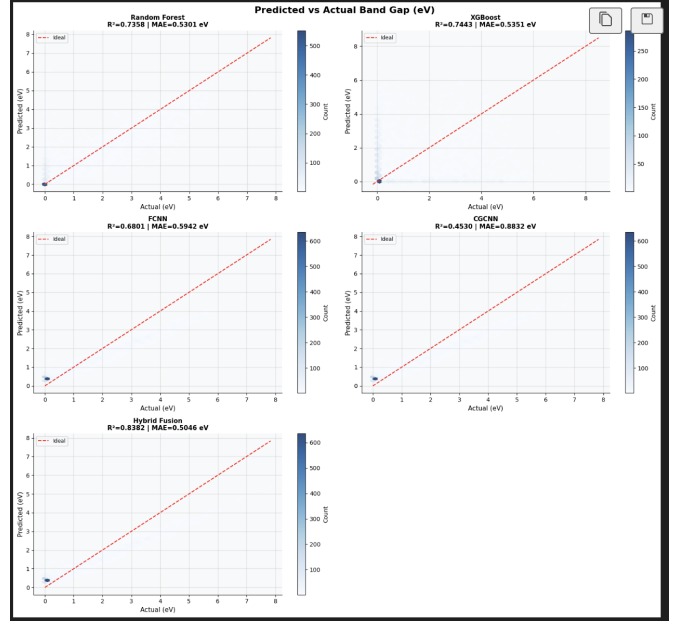


Fig. 2. Predicted vs Actual values for different models.

The predicted versus actual plots provide insights into how closely each model approximates the true values. Ideally, predictions should lie along the diagonal line, representing perfect prediction.

The **Hybrid Fusion model** demonstrates predictions that are closely aligned with the ideal diagonal line, indicating high accuracy and consistency. In contrast, other models exhibit greater dispersion around the diagonal, reflecting higher prediction errors.

The **Random Forest** and **XGBoost** models show moderate alignment with the ideal line, suggesting that they capture underlying trends effectively but still suffer from some deviation. The **FCNN** and **CGCNN** models show larger deviations, indicating weaker generalization performance.

C. Final Model Comparison

Final Model Results Summary			
Model	MAE (eV)	RMSE (eV)	R^2
Hybrid Fusion	0.5046	0.6341	0.8382
XGBoost	0.5351	0.8204	0.7443
Random Forest	0.5301	0.8339	0.7358
FCNN	0.5942	0.9176	0.6801
CGCNN	0.8832	1.1659	0.453

Fig. 3. Final summary of model performance.

The final results summary clearly highlights the superiority of the hybrid model. The integration of machine learning and deep learning predictions allows the model to benefit from both structured data learning and nonlinear pattern extraction.

D. Discussion

The results demonstrate that machine learning models such as XGBoost and Random Forest are highly effective for structured tabular data due to their ability to handle feature interactions efficiently. Deep learning models provide additional capability for capturing nonlinear relationships but require larger datasets for optimal performance.

The hybrid model successfully combines the advantages of both approaches, resulting in improved accuracy, reduced error, and better generalization. This indicates that integrating multiple learning paradigms is a promising direction for material property prediction.

Overall, the experimental results validate the effectiveness of the proposed methodology and highlight the importance of combining different modeling techniques to achieve superior performance.

IX Discussion

The experimental results demonstrate the effectiveness of the proposed approach for material property prediction using both machine learning and deep learning techniques. A detailed analysis of the results reveals that different models exhibit varying levels of performance depending on their ability to capture feature interactions and nonlinear relationships within the dataset.

The **Hybrid Fusion model** consistently outperforms all individual models across all evaluation metrics. It achieves the highest R^2 value, indicating strong correlation between predicted and actual values, while also maintaining the lowest MAE and RMSE values. This improvement can be attributed to the combination of structured learning from machine learning models and nonlinear representation learning from deep learning models. The hybrid approach effectively integrates these complementary strengths, resulting in improved prediction accuracy and robustness.

Among the individual models, **XGBoost** and **Random Forest** demonstrate strong performance due to their ability to handle structured tabular data efficiently. These models capture feature interactions through ensemble learning and provide stable predictions with relatively low error values. However, their performance is slightly limited compared to the hybrid model, as they primarily focus on structured relationships and may not fully capture complex nonlinear dependencies.

The **Feedforward Neural Network (FCNN)** shows moderate performance, indicating its ability to learn nonlinear relationships within the dataset. However, its performance is influenced by the size and nature of the dataset. Since deep learning models typically require large datasets for optimal performance, the moderate dataset size limits its ability to fully generalize complex patterns.

The **CGCNN model** exhibits the lowest performance among all models. This can be explained by the fact that CGCNN is specifically designed for graph-based material representations, whereas the current dataset is structured and tabular in nature. As a result, the model is unable to effectively capture relevant feature relationships, leading to higher prediction errors.

The predicted versus actual plots further support these observations. The hybrid model produces predictions that closely align with the ideal diagonal line, indicating high accuracy and consistency. In contrast, other models show varying degrees of deviation, reflecting differences in their ability to approximate the true values.

Overall, the results highlight the importance of selecting appropriate models based on data characteristics. Machine learning models are highly effective for structured data, while deep learning models are better suited for capturing nonlinear patterns. The integration of both approaches through a hybrid model provides a balanced and efficient solution for material property prediction.

This study demonstrates that combining multiple learning techniques can significantly improve performance and offers a scalable and reliable approach for real-world engineering applications.

X Conclusion

This study presents a comprehensive approach for material property prediction using machine learning, deep learning, and hybrid modeling techniques. The objective was to evaluate the effectiveness of different models in predicting concrete compressive strength and to identify the most suitable approach for handling structured tabular data with complex nonlinear relationships.

The experimental results demonstrate that machine learning models such as XGBoost and Random Forest provide strong baseline performance due to their ability to efficiently capture feature interactions in structured data. Deep learning models, particularly the Feedforward Neural Network, further enhance the ability to model nonlinear dependencies, although their performance is influenced by dataset size and complexity.

The proposed **Hybrid Fusion model** achieves the best overall performance by combining the strengths of both machine learning and deep learning approaches. It produces higher prediction accuracy, lower error values, and better generalization compared to individual models. This confirms that integrating multiple learning paradigms can effectively improve model performance for material property prediction tasks.

Furthermore, the analysis highlights the importance of selecting appropriate models based on data characteristics. While traditional machine learning models are well-suited for tabular datasets, deep learning models contribute additional capability in capturing complex patterns. The hybrid approach successfully balances these aspects, making it a robust and scalable solution.

In conclusion, the proposed methodology provides an efficient and accurate framework for predicting material properties, reducing reliance on time-consuming experimental methods. Future work can focus on extending this approach to larger datasets, incorporating additional material features, and exploring advanced deep learning architectures to further enhance prediction performance.

XI Future Work

While the proposed framework demonstrates strong performance for material property prediction, several directions can be explored to further enhance its effectiveness and applicability. One potential improvement is the use of larger and more diverse datasets, which would enable deep learning models to better capture complex nonlinear relationships and improve generalization across different material compositions.

Future work can also focus on incorporating additional features such as environmental conditions, temperature variations, and material processing parameters, which may have a significant impact on material properties. The inclusion of such factors can lead to more accurate and realistic predictions in practical scenarios.

Another promising direction is the exploration of advanced deep learning architectures, including graph-based models and attention mechanisms, which can better capture structural and relational information within materials. These approaches may provide improved performance, especially for datasets with complex dependencies.

Furthermore, the hybrid fusion strategy can be enhanced by adopting adaptive weighting mechanisms instead of fixed weights, allowing the model to dynamically adjust the contribution of machine learning and deep learning components based on data characteristics.

Finally, the proposed framework can be extended to other material science problems, including prediction of thermal, electrical, and mechanical properties across different domains. This would demonstrate the scalability and versatility of the approach in real-world engineering applications.

References

- [1] H. W. Kuhn and A. W. Tucker, "Nonlinear programming," in *Proc. 2nd Berkeley Symp. Mathematical Statistics and Probability*, 1951.
- [2] R. Caruana and A. Niculescu-Mizil, "An empirical comparison of supervised learning algorithms," in *Proc. ICML*, 2006.
- [3] Y. Bengio, "Learning deep architectures for AI," *Foundations and Trends in Machine Learning*, vol. 2, no. 1, pp. 1–127, 2009.
- [4] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, 2012.
- [5] G. E. Hinton, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.
- [6] A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*, 2012.
- [7] S. Ruder, "An overview of gradient descent optimization algorithms," arXiv preprint arXiv:1609.04747, 2016.
- [8] X. He, K. Zhao, and X. Chu, "AutoML: A survey of the state-of-the-art," *Knowledge-Based Systems*, vol. 212, 2021.
- [9] J. Schmidhuber, "Deep learning in neural networks: An overview," *Neural Networks*, vol. 61, pp. 85–117, 2015.
- [10] M. Abadi et al., "TensorFlow: A system for large-scale machine learning," in *Proc. OSDI*, 2016.